

Network Intrusion Detection Combining Laplacian Eigenmaps and Enhanced Density Peak Clustering Algorithms

Aihua Nian¹ and Peng Zhang²

(Corresponding author: Peng Zhang)

Institute of Technology, Lanzhou Modern Vocational College, Lanzhou 730300, China¹
Network and Computing Center, Shenyang Institute of Engineering, Shenyang 110136, China²

Email: zhangpeng200312@126.com

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Abstract

Laplacian Eigenmaps (LE) are combined with an enhanced Density Peak Clustering (DPC) algorithm to improve the strength of network intrusion prevention detection. Using the quartile and standard deviation to set the threshold and using LE for dimensionality reduction, the results of testing LE-DPC showed that high accuracy and efficiency were achieved on UCI datasets, among which the accuracy was 0.930 on the KDD-Cup 1999 dataset. The system effectively maintained cluster quality and reduced misclassification. After dimensionality reduction, the LE-DPC system continued to maintain good cluster quality, further reducing the number of misclassified points. Overall, the study confirms the applicability and advantages of the LE-DPC algorithm in network intrusion detection scenarios, especially in high-dimensional complex data environments. It provides an effective real-time network security defense solution in high-dimensional data environments.

Keywords: Cluster Analysis; Density Peak Clustering; High-dimensional Data Reduction; Laplacian Eigenmaps; Network Intrusion Detection

1 Introduction

Network security is a major challenge in the information society. Network intrusion detection is an essential defense mechanism and occupies a central position in research on network security. Although security technologies such as bastion hosts, firewalls, and intrusion detection systems are widely used for daily network protection, the emergence of Advanced Persistent Threats (APTs) and new network attack patterns, such as internal threats, pose significant challenges to existing network defenses. The evolution of network defense techniques, from firewall-based perimeter protection to multidimensional intrusion detection, is continuously progressing and

improving [4, 11, 17]. Network intrusion detection, originating in the 1980s, has gone through stages of statistical detection, rule-based detection, and behavior-based detection, forming a comprehensive theoretical framework and practical methods. However, the continuous expansion of network scale and increasing complexity of network behavior have made network environments and attack methods more diverse and stealthy, posing severe challenges to existing Network Intrusion Detection Systems (NIDS) [2, 12, 18]. In recent years, numerous researchers have delved into the study and enhancement of network intrusion detection methods using machine learning and data mining techniques, especially clustering analysis. These methods aim to reveal hidden patterns and regularities within large-scale network traffic data, providing effective references for identifying network attacks and abnormal behaviors. They exhibit significant advantages in handling large-scale, high-dimensional, and complex network traffic data. However, many existing clustering analysis methods construct models based on the global properties of data distribution. For local behaviors within the network environment and small-scale but high-risk attack behaviors, the detection effectiveness of such methods is often unsatisfactory. Furthermore, the efficiency of existing clustering analysis methods in addressing practical problems is impacted by various factors, including the choice of data preprocessing and cluster center initialization methods, the convergence and stability of the iterative process, and the level of computational resource requirements [6, 8, 16].

The evolution of NIDS has gone through many stages, including statistical detection, rule-based detection and behavior-based detection. Statistical detection methods rely on the statistical characteristics of network traffic to identify abnormal behaviors. Such methods are simple but easy to be evaded by attackers. Rule-based detection systems use known attack signatures to identify intrusion attempts. While they perform well in detecting known

attack types, they are less effective against unknown or variant attacks. Behavior-based detection methods identify potential threats by analyzing deviations in user or system behavior. These methods are flexible and adaptable to new attack patterns, but they require extensive data analysis and processing.

Network security and its intrusion detection are major challenges in the field of information technology. With the development of Internet technology and the increasing complexity of network environment, network security threats are constantly evolving, including APTs, internal threats and various new network attacks. Network security protection faces unprecedented challenges due to the rapidly changing attack methods of highly hidden and technologically complex threats. The development of efficient and accurate network intrusion detection technology is vital for ensuring network security and stable operation of information systems. This technology can detect and respond to security threats in a timely manner.

Against this backdrop, this study focuses on an analysis approach that combines the selection of adaptive optimal cluster centers with sensitivity to local characteristics. It thoroughly investigates a network intrusion detection method that integrates Laplacian Eigenmaps (LE) and an improved Density Peak Clustering (DPC) algorithm. The combination of LE and DPC is expected not only to enhance the precision of network intrusion detection but also to ensure performance and effectiveness in large-scale network environments, providing robust support for real-world network security protection efforts.

The first part of this study analyzes the global and domestic forms of network security and network security detection. The second part focuses on designing a NIDS that combines the LE algorithm with the improved DPC algorithm. The algorithmic process and parameter settings are detailed, which is an innovative aspect of this study. The third part involves a detailed analysis of the performance of the LE-DPC algorithm. Through systematic performance simulation analysis, it is demonstrated that the LE-DPC algorithm exhibits high accuracy and efficiency across all test datasets. The LE-DPC system's application in intrusion detection and its comparison with other common algorithms are thoroughly analyzed. The fourth part analyzes the experimental results obtained and summarizes the shortcomings of the research.

2 Related Works

Network security has consistently been a focal point in the field of information technology, with the rapid development of NIDS. In recent years, as cyber attack methods have evolved and the network environment has become increasingly complex, traditional intrusion detection technologies based on signatures and anomalies have faced growing challenges. To enhance detection accuracy and efficiency, many researchers have started exploring the application of machine learning methods to

intrusion detection. The team led by Ma proposed an Adversarial Domain Adaptation (ADDA) approach employing a dual-domain alignment strategy for Internet of Things (IoT) intrusion detection with limited samples. They combined the dual-domain alignment strategy with the ADDA model for supervised domain adaptation. A lightweight network was designed to extract features, along with the development of new loss functions and domain discriminator models. The research results indicated the effectiveness of the model in detecting intrusions in a minority of samples from IoT devices [3]. The team led by Mu introduced a method for vehicular intrusion detection using Sparse Regularized Convolutional Auto-encoders (SRCAE) and flow clustering. The intrusion detection was achieved by describing the profile baseline of normal communication messages through the model. Tests demonstrated that the model exhibited good performance and real-time capabilities in anomaly detection, with a detection accuracy of 98.43% for unknown attacks and an F1 score of 97.15% [20]. The team led by J Li proposed a deep learning framework called VNN, which is a voting-based approach that integrates different deep learning models to analyze various aspects of the data. By aggregating multiple models, the framework aimed to achieve more accurate and robust detection capabilities, particularly enhancing the accuracy of complex attack detection. Experimental validation using the KDDCUP'99 and CTU-13 datasets demonstrated that the voting mechanism of VNN significantly improved system performance, reducing false positives by up to 75% compared to the original deep learning models [5].

Cluster algorithms show potential in detecting unknown attacks due to their effectiveness on unlabeled data. However, traditional clustering algorithms often encounter difficulties when applied directly to high-dimensional, large-scale datasets, which limits their practical use in network intrusion detection. The application of clustering algorithms is widespread, and the JN's team proposed a mechanism to maximize the efficiency of Heterogeneous Networks (HetNets) by improving the k-means clustering and dynamic load-based Small Cell (SC) switching algorithm. This clustering algorithm optimized initial centroids to achieve maximum inter-cluster separation. Simulation results indicated a performance improvement of 20% in energy efficiency and 16% in system throughput compared to traditional k-means clustering methods [13]. EGavags developed a novel K-nearest Neighbor Classification Method (KNN-CCL) that divides the training dataset into multiple parts using a parallel centroid-based hierarchical clustering algorithm. A four-stage refinement process was employed to generate high-quality clusters. Experimental verification on the UCI dataset demonstrated the superiority of KNN-CCL in terms of classification accuracy and performance [7]. The team led by ZYang proposed a noise-robust Intuitive Fuzzy C-Means (NR-IFCM) algorithm. This algorithm addressed noise and data uncertainty issues by introducing a noise-robust Intuitive Fuzzy Set (NR-IFS) and a new

intuitive fuzzy factor combining local grayscale and spatial information. The NR-IFCM algorithm was applied to process human brain MRI with noise. Experimental results on a significant number of human brain MRI experiments confirmed the effectiveness and superiority of the proposed algorithm [10].

In summary, the challenge in intrusion detection systems is to improve the accuracy and efficiency of network intrusion detection in high-dimensional, large-scale data environments. Therefore, combining LE algorithm and improved DPC presents a new solution for efficient and accurate network intrusion detection. The study will focus on analyzing and discussing the application prospects of these two algorithms in the field of intrusion detection and the potential advantages of their combination.

3 Building a Network Intrusion Detection System with LE-Optimized DPC Algorithm

This research delves into the construction of an advanced NIDS by combining the LE algorithm with an improved DPC algorithm. This text explains the theoretical foundation and technical details of the LE-optimized DPC algorithm. It also describes how the algorithm's capability to handle high-dimensional data is enhanced through feature reduction and automatic determination of cluster centers. Subsequently, it demonstrates the application of the optimized algorithm in a NIDS, providing detailed insights into the conceptualization of system design, feature planning, and the integration and execution of the algorithm in real-world workflows.

3.1 LE-Optimized DPC Algorithm

The DPC algorithm possesses the advantage of identifying arbitrary clusters, but it faces challenges such as manual adjustment of cluster centers and unclear clustering in high-dimensional data. Therefore, the paper focuses on optimizing the automatic cluster center determination strategy for the DPC algorithm. A new threshold is constructed using the inter-quartile range and standard deviation after normalizing the dataset D , which is then employed to automatically confirm initial cluster centers. The DPC algorithm structure as shown in Figure 1.

After normalizing the dataset D , the algorithm initiates an iterative process, traversing pairs of data points and calculating the distances between them. Subsequently, the algorithm computes the local density of each data point. In this step, the algorithm seeks cluster structures within the data based on the compactness or concentration within these clusters, defining their respective densities. For data points with high similarity, their local densities should be close, aiding the algorithm in identifying an independent cluster. Next, for each non-highest density data point, the algorithm calculates its minimum

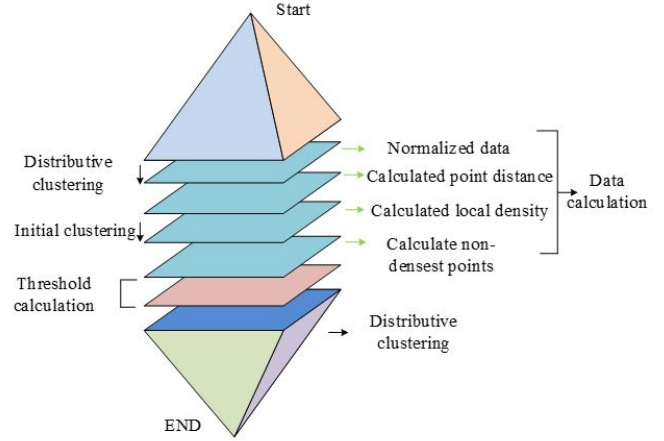


Figure 1: A DPC algorithm flow

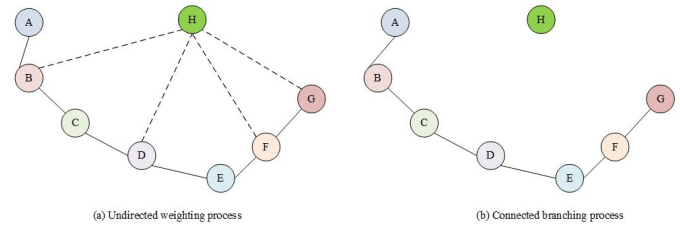


Figure 2: Cluster Class Merging Process *** The figure havenot cited in this paper. ***

distance to higher density points and selects its maximum distance to other points. Additionally, the algorithm computes a threshold based on these minimum and maximum distances, using it to choose initial cluster centers. These cluster centers represent high-density regions in the data space, providing anchor points for distance comparisons and cluster assignments for other points. After this step, the algorithm assigns each non-initial cluster center point to the nearest high-density cluster center. The process is completed by utilizing the connectivity relationships to construct an undirected weighted graph that fully represents the structure and relationships within the data. The Cluster Class Merging Process as shown in Figure 2.

The concepts of independent connected branches and cut edges are used to further refine clustering results. In the end, the algorithm removes these cut edges, causing the connected branches in the graph to be separated. These connected branches are precisely the clusters that the algorithm aims to identify, composed of a group of points with similar density, forming a relatively dense region in the data space. Ultimately, these connected branches or clusters are output by the algorithm as the final clustering results. Therefore, the fundamental optimization of the DPC algorithm enhances its clustering effectiveness, making it more suitable for handling complex datasets. In the basic optimization algorithm, local

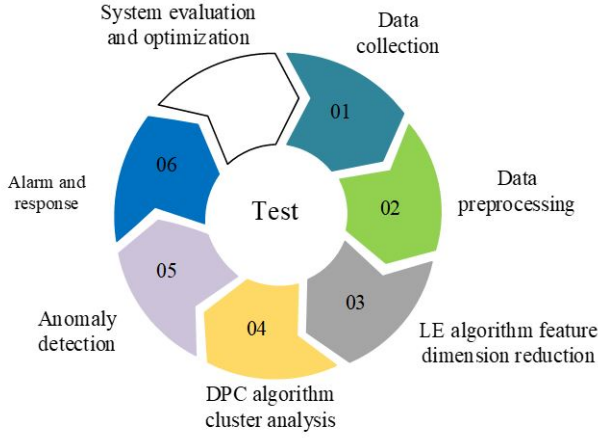


Figure 3: Network Detection System Flow

density calculation is represented by Formula (1).

$$\rho_i = \sum_{i \neq j} \chi(d_{ij} - d_c) \quad (1)$$

The use of kernel distance method to calculate local density can be expressed by Formula (2).

$$\rho_i = \sum_{i \neq j} \exp \left[- \left(\frac{d_{ij}}{d_c} \right)^2 \right] \quad (2)$$

In Formulas (1) and (2), d_{ij} represents the Euclidean distance, d_c is the cutoff distance, and i denotes the data point. Thus, faced with the limitation of manually selecting cluster centers in DPC analysis, the algorithm automates the process of determining the core nodes of clusters. The primary step involves calculating the product of each data point's local density and its distance to high-density data points. Subsequently, statistical methods, including quartiles and standard deviation, are introduced to define a filtering threshold. Points with larger products are then selected as preliminary cluster cores using this threshold. Furthermore, a weighted undirected graph is constructed based on these preliminary core points, where the edge weights represent the strength of connections between clusters. By analyzing the graph structure, disconnected connected branches are identified, and the corresponding clusters are merged to generate the final clustering structure. However, due to the unclear clustering effects in high-dimensional datasets, the introduction of LE is applied to optimize the DPC algorithm [1, 9, 14, 15, 19, 21].

3.2 Construction of NIDS

After describing the optimization process based on the LE and the improved DPC algorithm, this study further integrates the optimized algorithms into the NIDS, as shown in Figure 3.

As illustrated in Figure 3, the constructed NIDS framework starts with meticulous data preprocessing, aiming

to reduce noise and prepare data for subsequent algorithmic processing. Subsequently, the LE algorithm implements dimensionality reduction in the feature space, striving to simplify complexity while maintaining data integrity. The optimized DPC algorithm then intervenes, analyzing the reduced-dimensional data using advanced clustering techniques to identify abnormal patterns. The system ultimately presents easily interpretable detection results through a series of precise decisions, while monitoring system performance through a cyclical evaluation mechanism to ensure its adaptability to rapidly changing network environments. In the LE-DPC system, local information between data points is preserved through a k-nearest neighbors graph, where larger weights indicate closer proximity, signifying a more pronounced clustering effect. When constructing the nearest neighbors graph, the weight function calculation formula is given in Formula (3).

$$W_{ij} = \exp \left(- \frac{\|x_i - x_j\|^2}{t} \right) \quad (3)$$

In Formula (3), t is the scaling factor for adjusting the Gaussian and function, and x_i and x_j represent two different data points in the dataset. To address the issue of LE introducing parameters that cannot be automatically input into the DPC model, a global neighborhood radius is introduced into the algorithm, as shown in Formula (4).

$$d_m = \frac{1}{n-1} \sum_{j=1, i \neq j}^n d_{ij} \quad (4)$$

In Formula (4), n is the summary of points in the dataset, d_{ij} is the distance between two points, and d_m is the global distance radius. The number of neighbors can be calculated using Formula (5).

$$k = \frac{1}{n-1} \sum_{i=1}^n X(d_{ij} - d_m) \quad (5)$$

In Formula (5), $X()$ is the number of other points in the neighborhood radius corresponding to the point, and the local scale parameter can be calculated using Formula (6).

$$\sigma_i = \sqrt{\frac{1}{n-1} \sum_{j=1}^{D_i} (d_{ij} - d_m)^2} \quad (6)$$

In Formula (6), σ is the local scale parameter, and D_i is the number of samples in the neighborhood radius corresponding to the point. The weight calculation formula of W'_{ij} is given in Formula (7).

$$W'_{ij} = \left(\exp - \frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j} \right) \quad (7)$$

Subsequently, the Laplace matrix is calculated using the weights of equation (7), and the minimum eigenvalue of the Laplace matrix corresponding to the low dimensional

mapping of the high-dimensional data is obtained. In the dimensionality reduction process, smaller weights indicate greater distance between data points, while larger weights indicate closer proximity, ensuring that the original structure of the data is not lost during dimensionality reduction. The distance for the reduced data can be calculated using Formula (8).

$$(8)d_{ij}^{(k)} = \sqrt{\sum_{m=1}^k (x_{im} - x_{jm})^2} \quad (8)$$

In Formula (8), $d_{ij}^{(k)}$ represents the distance between two points, and x_{im} and x_{jm} denote the coordinates of the corresponding points in the respective dimensions. After obtaining the corresponding data representation in the low-dimensional space, it is necessary to use the optimized DPC algorithm for clustering. First, the local density of each data point in the low-dimensional space needs to be calculated. Then, the clustering center for each data point is determined by calculating the distance to high-density data points. For the data point with the highest local density, its distance is defined as the maximum distance to all data points. The formula for calculating the clustering center is shown in Formula (9).

$$\rho_i = \sum_{j \neq i}^n \chi(d_{ij}^{(k)} - \delta) \quad (9)$$

In Formula (9), δ is a preset threshold. With these preliminary clustering centers, an undirected weighted graph based on these clustering centers can be constructed. The weight of the edges in the graph is defined as the distance between two data points, and the formula for calculating the weight is shown in Formula (10).

$$w_{ij} = \exp\left(-\frac{(d_{ij}^{(k)})^2}{2\sigma^2}\right) \quad (10)$$

In Formula (10), w_{ij} is the weight of the edge between two points, and σ is a parameter measuring the distance scale. During the experimental process, parameters such as Gaussian radius, number of neighbors, etc., need to be adjusted, as these parameters have a significant impact on the clustering results. For the calculation of the Gaussian radius, Formula (11) is used.

$$\sigma = \frac{\sum_j^i d_{ij}^{(k)}}{N(N-1)} \quad (11)$$

σ is the Gaussian radius calculated from the average distance of all point pairs, used for weight calculation. N is the total number of points in the dataset. The number of neighbors is defined by its calculation in Formula (12).

$$N_c = \left| \{j \geq \left| d_{ij}^{(k)} < \tau \right| \} \right| \quad (12)$$

In Formula (12), N_c is the number of neighbors for a point, i.e., the number of points with distances less than τ , where

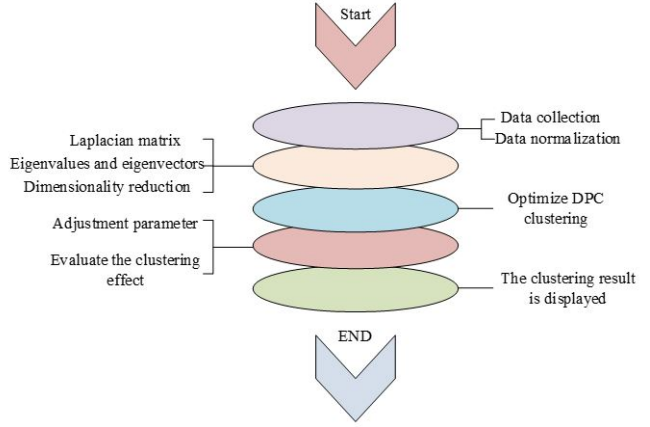


Figure 4: Optimization Algorithm Flow

TE is a preset distance threshold. Finally, clustering can be performed based on Formula (9) and Formula (10). The construction process is illustrated in Figure 4.

The Laplacian matrix construction formula is represented by Formula (13).

$$L = D - W \quad (13)$$

In Formula (13), D is the degree matrix, and W is the adjacency matrix. The new threshold calculation formula can be expressed by Formula (14).

$$T = Q_3 + k \cdot IQR \quad (14)$$

In Formula (14), T is the calculated threshold, Q_3 is the third quartile of the dataset, i.e., the value at the 75th percentile when the dataset is sorted in ascending order. k is a constant factor used to adjust the strictness of outlier identification. IQR is the interquartile range. In the calculation of Laplacian eigenmapping for feature vector calculation, Formula (15) can be used.

$$\Delta v = \lambda L v \quad (15)$$

In Formula (15), Δ is the eigenvalue matrix of the Laplacian matrix containing all eigenvalues. v is the corresponding eigenvector matrix. λ is a specific eigenvalue, and L is the Laplacian matrix. In the algorithmic experiments, the adjusted Rand index and normalized mutual information are used to measure classification accuracy. The formula for the adjusted Rand index is shown in Formula (16).

$$ARI = \frac{RI - E(RI)}{\max(RI) - E(RI)} \quad (16)$$

In Formula (16), RI is the Rand index, and $E(RI)$ is a mathematical expectation between -1 and 1.

$$NMI = \frac{MI(\hat{C}, C)}{\max(H(\hat{C}), H(C))} \quad (17)$$

In Formula (17), NMI is the normalized mutual information. $MI(\hat{C}, C)$ is the value of the true result compared

Table 1: Software and Hardware Environments

Hardware and software composition	Instructions
Operating system	Windows 10 64 bit
CPU	Intel i5 7th Generation
Internal memory	8GB
Hard disk	1TB HDD
Programming language	Python
Data processing tool	Numpy, Pandas
Data visualization tool	Matplotlib
Machine learning library	Scikit-learn

to the clustering result. $H(\hat{C})$ is the information entropy of the true result, and $H(C)$ is the information entropy of the clustering result.

4 Application and Analysis of the LE-DPC Optimized System

The research was conducted to validate the effectiveness of the LE-DPC optimized system in network intrusion detection through systematic performance simulation and practical application analysis. The system's superior performance, including accuracy, detection speed, and false positive rate, was demonstrated across various test datasets compared to other commonly used algorithms. Ultimately, these detailed evaluations confirmed the practical value of the LE-DPC system in addressing real-world network security challenges.

4.1 System Performance Simulation Analysis

This text analyzed the performance simulation of a system, discussing the impact of different operations on its performance and how to optimize it by adjusting parameters. In the experimental environment setup, different software and hardware environments are described in Table 1.

The software and hardware configurations are as shown in Table 1, with Numpy and Pandas selected as data processing tools. Various publicly available UCI datasets were chosen for experiments, and the dataset descriptions are provided in Table 2.

The KDD-Cup 1999 and NSL-KDD datasets have the same number of features but differ in the number of instances, with NSL-KDD having less redundant data. The CIC-IDS 2017 dataset's feature and instance counts vary with dataset versions. The study initially selected feature gap sequences to assess algorithm performance on the datasets. The intrusion detection results are illustrated in Figure 5.

In the comparative study of network intrusion detection, Figure 5 reveals that the LA-DPC algorithm exhibits significant performance advantages across all datasets. When processing the KDD-Cup 1999 and NSL-KDD network intrusion datasets, the LE-DPC algorithm's fea-

ture gap sequences consistently demonstrated clear advantages. A smaller feature gap sequence indicated stronger capability in handling and distinguishing high-dimensional data. For all feature points in both KDD-Cup and NSL-KDD datasets, LE-DPC had the smallest feature gap sequences, highlighting its distinct superiority in handling such high-dimensional and complex data.

As shown in Figure 6, in the evaluation and comparison of the performance of network intrusion detection algorithms, the LA-DPC algorithm demonstrated significant advantages in all tests. For AMI, it primarily measured the similarity between different clusters. From Figure 6(a), the performance of LA-DPC on all four datasets was superior to other algorithms such as SVM-DPC, NN-DPC, GA-DPC, and RF-DPC, proving its superiority in maintaining similarity within the dataset. Correspondingly, for ARI, this metric was also used to quantify whether the predicted clustering is similar to the true clustering. According to Figure 6(b), LA-DPC outperformed the other four algorithms on all datasets. On the KDD-Cup 1999 dataset, LA-DPC achieved an ARI of 0.730, significantly higher than other algorithms, with the closest being RF-DPC with an ARI value of 0.612. On the NSL-KDD, ISCX 2012 IDS, and CIC-IDS 2017 datasets, LA-DPC's ARI reached 0.700, 0.655, and 0.715, respectively. These experimental results further confirmed the outstanding performance of the LA-DPC algorithm in network intrusion detection and clustering analysis, especially when dealing with complex, diverse, and high-dimensional datasets.

4.2 LE-DPC Optimization for Intrusion Detection Analysis

To ensure efficient detection while preserving local data features and reducing data dimensions, the LE-DPC algorithm also needs to maintain high detection efficiency and provide accurate classification results in large-scale complex data environments. Therefore, a performance analysis of the LE-DPC algorithm in handling network intrusion detection problems, including clustering accuracy, computational efficiency, and detection accuracy, is conducted and compared with other popular algorithms to validate its practical superiority and feasibility. Table 3 presents the evaluation metrics for the algorithm.

Table 3 compares the performance of five algorithms in network intrusion detection. The LE-DPC algorithm achieved the highest accuracy at 91.4%, a recall rate of 89.7%, and precision of 92.3%. Its F1-score reached 91.0%, indicating excellent balance. In terms of efficiency, LE-DPC had a training time of only 40 seconds, with the smallest model size at 10.0MB. Additionally, its false negative rate was only 2.8%, and the false positive rate dropped to 4.2%, both of which were superior to other algorithms.

In the field of network intrusion detection, the LE-DPC algorithm had demonstrated significant advantages against different types of attacks. Specifically, as shown in

Table 2: Data set Description

	Feature quantity	Feature quantity	Feature quantity	Feature quantity	Feature quantity
KDD-Cup 1999 Data	41	490 million	5	Normal, DoS, R2L, U2R, Probe	~700 MB
NSL-KDD	41	148517	5	Normal, DoS, R2L, U2R, Probe	~150 MB
ISCX 2012 IDS	-	-	-	Multiple types of network attacks	Unfixed
CIC-IDS 2017	80+	2.7 million	Multiple network attacks	DDoS, DoS, Brute Force, Infiltration	~12 GB

Table 3: Index Assessment

Algorithm	Accuracy (%)	Recall (%)	Precision (%)	F1-score (%)	Training Time (s)	Model Size (MB)	Miss Rate (%)	False Alarm Rate (%)
DPC	84.5	80.2	85.3	82.7	60	15.2	4.2	8
RDP	86	83.5	86.9	85.1	70	20.8	3.9	7
SVM	87.2	85.6	88.1	86.8	80	30	3.6	6.8
Multi-CNN	89	87.3	90.1	88.6	120	50.5	3.1	5.7
LA-DPC	91.4	89.7	92.3	91	40	10	2.8	4.2

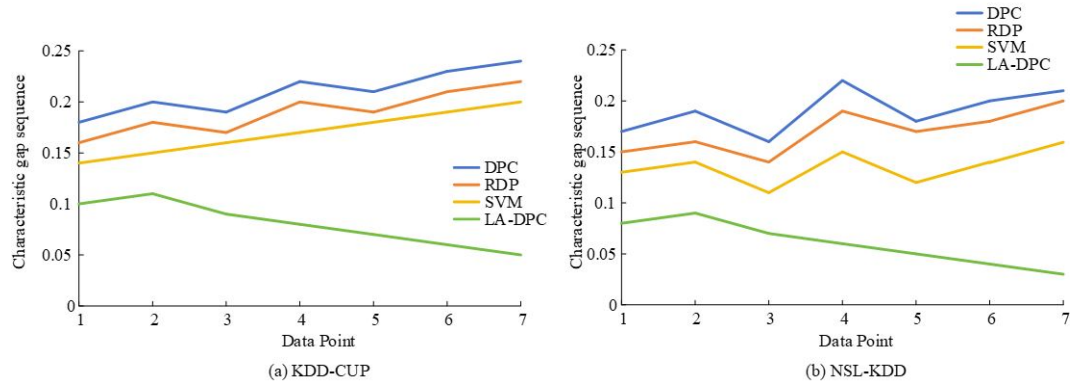


Figure 5: Feature Gap Sequence Comparison

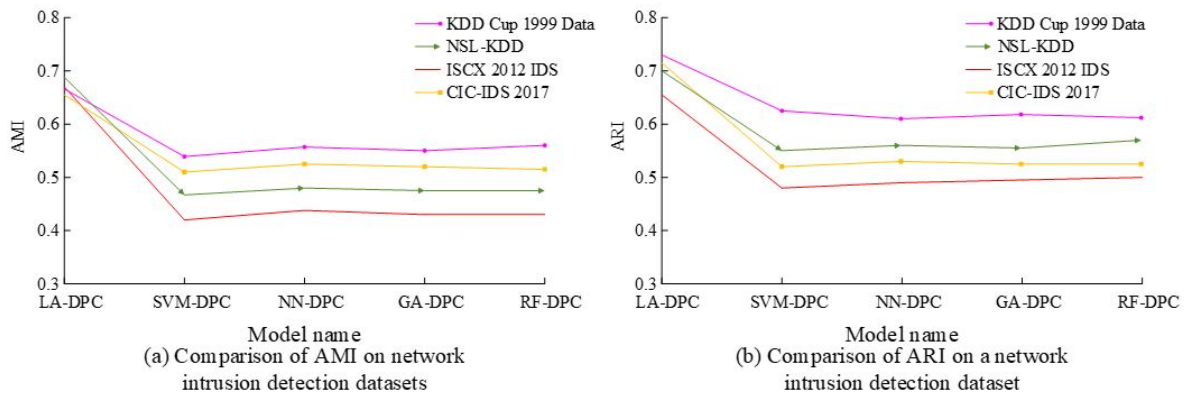


Figure 6: Comparison of ARI and AMI

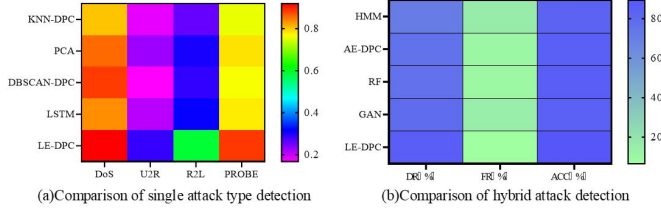


Figure 7: Comparison of Attack Type Detection

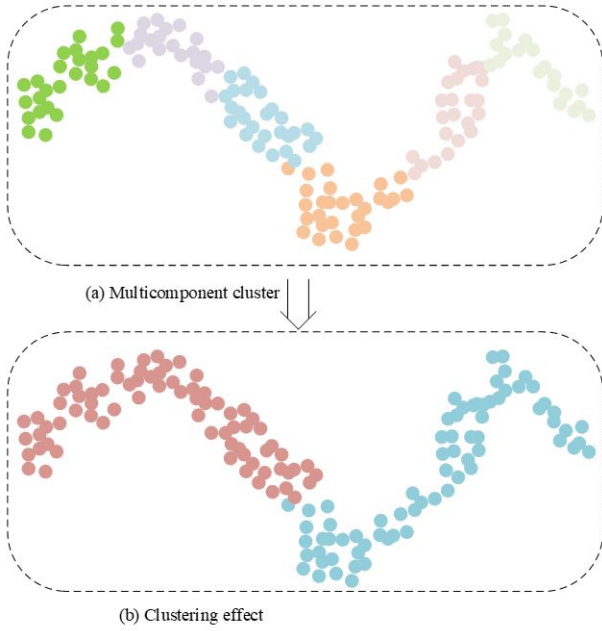


Figure 8: Display of Clustering Effect of Algorithm

Figure 7(a), the recognition rate of LE-DPC for DoS attacks was 91.8%, which was much higher than other algorithms, showing its excellent capability in handling large-scale service request attacks. Additionally, in the identification of U2R attacks, LE-DPC led other algorithms with a recognition rate of 28.5%. Figure 7(b) presents a comparison of mixed attack detection, where LE-DPC algorithm exhibits remarkable performance advantages as evidenced by the detection rate (DR), false positive rate (FR), and overall accuracy (ACC) compared to NIDS. Compared to HMM, AE-DPC, RF, and GAN algorithms, LE-DPC achieved the highest score of 85.7% in DR, indicating its precise capture and correct classification of the majority of intrusion behaviors. Furthermore, while reducing the FR, LE-DPC maintained an FR of only 6.5%, revealing its ability to balance high defensive performance in complex network environments with ideal efficiency and accuracy.

From the algorithm clustering results in Figure 8, it can be observed that the research algorithm effectively classifies points in clusters, with almost no misjudgment of cluster points as noise points. The clustering effect is evident, accurately dividing the data into two categories.

Table 4: Number of False Points

Data set	Data point	DBSCAN	NJW	kMeans-DPC	SVM-BP	LE-DPC
Pathbased	300	52	63	59	18	2
ThreeCircles	3603	25	755	47	0	0
Strip	400	7	8	12	1	0
Aggregation	788	43	74	123	6	3
Compound	399	15	35	38	2	1
D31	3100	90	120	150	4	2
Flame	240	30	40	55	9	1

Table 4 compares the number of misclassified points, with the research using this key metric to measure the handling capability of different algorithms on the dataset. Table 4 shows the number of misclassified points for six clustering algorithms on seven different datasets. The datasets represent diverse data distributions and clustering challenges, ranging from simple two-dimensional datasets (e.g., Flame) to high-dimensional datasets with complex pattern features (e.g., D31 and ThreeCircles). From Table 3, the LE-DPC algorithm had a significant performance advantage, consistently exhibiting the fewest misclassified points. For example, on the ThreeCircles dataset, both LE-DPC and SVM-BP had no misclassifications, while other algorithms had a minimum of 25 misclassified points. LE-DPC maintained low misclassification points while demonstrating stable clustering effects across all datasets. Table 5 provides a comparison of algorithms under different evaluation metrics.

Table 5 evaluates the LE-DPC algorithm and four other clustering algorithms using five different metrics, including clustering purity, silhouette coefficient, Davies-Bouldin index, mutual information score, and execution time. The LE-DPC algorithm consistently outperformed others across all evaluation metrics. It maintained high clustering purity, low Davies-Bouldin score, reflecting excellent internal cluster cohesion and distinctiveness from other clusters. Additionally, LE-DPC achieved a mutual information score of 0.75, confirming its accuracy in capturing the intrinsic structure of the data. Moreover, LE-DPC demonstrated efficiency, with lower execution time compared to other algorithms, highlighting its potential for real-time and efficient clustering demands. In summary, LE-DPC exhibited robust performance in complex and challenging clustering tasks, showing versatility and broad application prospects.

Figure 9 shows the comparison of accuracy rates of different algorithms. The LE-DPC algorithm was superior to other four algorithms in each iteration stage. When the number of iterations was 20, the accuracy rate of LE-DPC was 85%. As the number of iterations increased, the accuracy rate gradually increased to 98%. In contrast, the initial accuracy rate of other algorithms was low and their growth rate was slow. It demonstrated the efficiency and reliability of LE-DPC in dealing with complex network security threats.

Table 5: Performance Comparison of Modern Clustering Algorithms with Multiple Indexes

Test direction	Purity (%)	Silhouette Coefficient	DB Index	Mutual Information Score	Execution Time (s)
Spectral-Cluster	80.2	0.42	1.24	0.68	45.5
Mean-Shift	78	0.38	1.35	0.62	25.3
Affinity-Propagation	76.4	0.35	1.5	0.59	52.7
Birch	79.9	0.4	1.3	0.65	55.7
LE-DPC	87.6	0.55	0.95	0.75	42.9

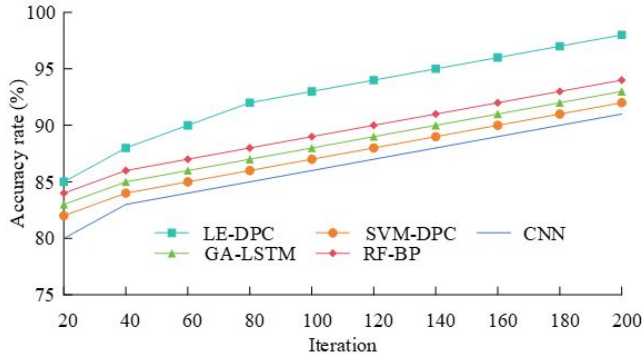


Figure 9: Comparison of the Accuracy of Different Algorithms

5 Conclusion

Currently, the severity of cyber security issues is increasing, and network intrusion detection has become a crucial tool for security protection. Intrusion detection systems face challenges in handling high-dimensional, large-scale, and complex network data. Enhancing their detection efficiency and accuracy has become a significant concern. The paper proposed a network intrusion detection strategy that combines LE with an optimized DPC algorithm. The results indicated that the LE-DPC algorithm maintained efficient detection efficiency in large-scale complex data environments and provided accurate classification results. The improved LE-DPC model achieved a 93% accuracy on the KDD-Cup 1999 dataset, showing a relative improvement of 14.5% and 12% compared to other models such as SVM-DPC and NN-DPC, respectively. It also maintained a high accuracy of 89.5% on the NSL-KDD dataset, significantly outperforming traditional algorithms. Moreover, concerning FR and DR, LE-DPC controlled the FR within 6.5% and achieved a DR of 85.7%, demonstrating superior intrusion detection capabilities. In summary, the LE-DPC algorithm significantly enhanced the accuracy and efficiency of network intrusion detection. One limitation of the study is the need for further optimization of its processing speed and power consumption when handling large data flows. Although the LE-DPC algorithm shows advantages in many aspects, there are still some limitations in the research. First, current research focuses on publicly available data sets, which may not be able to fully cover all situations encountered in the complex and changeable network environment of the

real world. In addition, the computational efficiency and energy consumption of the algorithm in processing very large data sets have not been deeply discussed. Therefore, future work needs to consider more challenges in real-world environments, including dynamic changes in data, evolving attack patterns, and optimal allocation of computing resources.

Future research can enhance the algorithm's defense capability against various network attacks and improve its universality and robustness. Additionally, applying the LE-DPC algorithm to other fields, such as fraud detection and abnormal behavior analysis, can verify its effectiveness in different application scenarios.

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Biography

Aihua Nian obtained his master's degree in the field of Computer Technology from Lanzhou University in 2010. Presently, he is working as an Associate Professor at the Institute of Technology of Lanzhou Modern Vocational College. His areas of interest include machine learning, image processing, and information security.

Peng Zhang obtained his Master's degree in Computer Science from Northeastern University in 2007. Currently, he works at the Network Center of Shenyang Institute of Engineering, where he holds the position of Associate Professor. His main research areas include database theory and technology, distributed and parallel systems, artificial intelligence technology, and network information security. As a professional in the field of computer science, he possesses extensive knowledge and profound research experience. He is dedicated to promoting technological innovation and application, having led multiple provincial and municipal-level projects, published numerous academic papers, and achieved significant research outcomes.